

Week 4: Fundamentals of Probability and Statistics II

PLSC 309 Quantitative Political Analysis

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September 17, 2026

AI Policy and Midterm Exam

AI Policy

- Use AI as a **learning tool** to help you understand course concepts.
- Do not use AI as an agent to complete your **homework** for you.

Midterm Exam

- **One hour, in person, open book, handwritten.**
- **40 questions** covering **Weeks 1–7**.
- Allowed materials: **printed slides and printed or handwritten notes**.
- **No electronic devices.**

Time Management

- 20 short questions: aim for **no more than 30 seconds each**.
- 20 longer questions: aim for **2 minutes each**.
- This leaves about **10 minutes to review**.
- No extra time. You will not have enough time to look up the answer to every question.

What Did We Learn Last Time and Last Last Time?

Sep 10th

- Statistical Inference
- Estimand, Estimator, Estimate
- Random sampling
- Independent, Identically Distributed (IID)
- Sampling distribution

Sep 15th

- Properties of Estimator
- Unbiasedness
- Efficiency
- Variance of the Sample Mean
- Standard Error
- Mean Squared Error
- Confidence Interval

Review: Confidence Interval

By Law of Large Number (consistency) and Central Limit Theorem (asymptotic normality):

$$\hat{\theta}_n \approx N\left(\theta, SE(\hat{\theta}_n)^2\right)$$

Standardizing:

$$Z = \frac{\hat{\theta}_n - \theta}{SE(\hat{\theta}_n)} \approx N(0, 1)$$

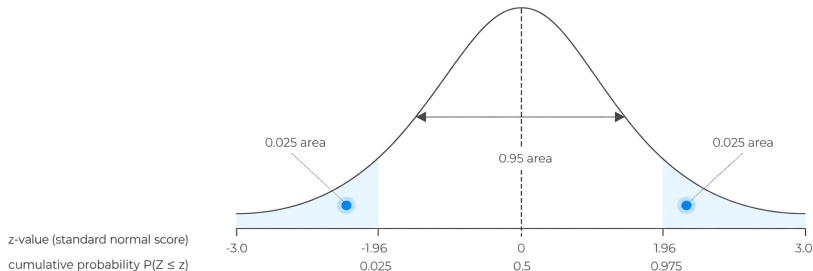
The confidence interval framework:

$$\hat{\theta}_n \pm \underbrace{z^*}_{\text{Critical Value}} \times \underbrace{SE(\hat{\theta}_n)}_{\text{Standard Error}}$$

Z marks how far you must go out from a sample estimate to capture a chosen percentage of uncertainty (such as 90%, 95%, 99%).

Review: Z-score/Z-value (95% CI)

$$\bar{Y}_n \pm 1.96 \times SE$$



```
qnorm(0.025, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] -1.959964
```

```
qnorm(0.975, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] 1.959964
```

Example: Pennsylvania Poll

Suppose we survey $n = 1,000$ Pennsylvania voters:

$$\hat{p} = 0.462$$

For a sample proportion:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.462(0.538)}{1000}} \approx 0.0158$$

The 95% confidence interval is:

$$\hat{p} \pm 1.96SE(\hat{p})$$

$$0.462 \pm 1.96(0.0158)$$

$$[0.431, 0.493]$$

Confidence Intervals from Repeated Samples

Suppose the true proportion of Pennsylvania voters who support the candidate is

$$p = 0.40.$$

We repeatedly draw random samples of $n = 1,000$ voters from this population.

For each sample, we calculate:

Point Estimate $\longrightarrow$ Standard Error $\longrightarrow$ 95% Confidence Interval.

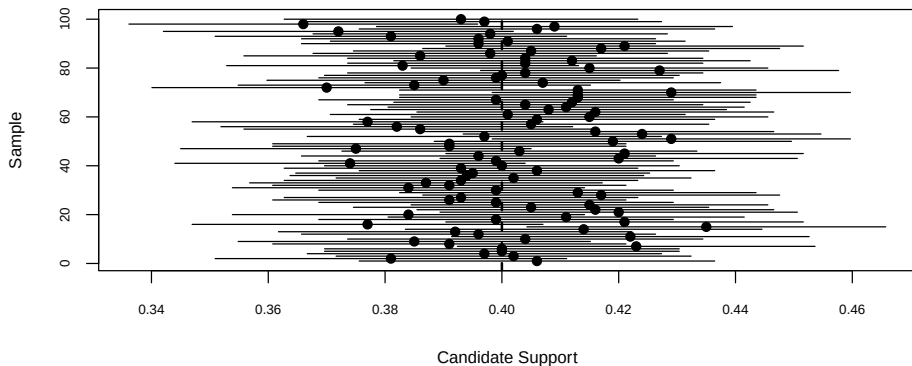
We repeat this process **100 times**.

What If We Repeat This 100 Times?

Now imagine repeating exactly the same procedure **100 times**.

Each random sample produces a different point estimate and therefore a different confidence interval.

95% Confidence Intervals from Repeated Samples (100 times)



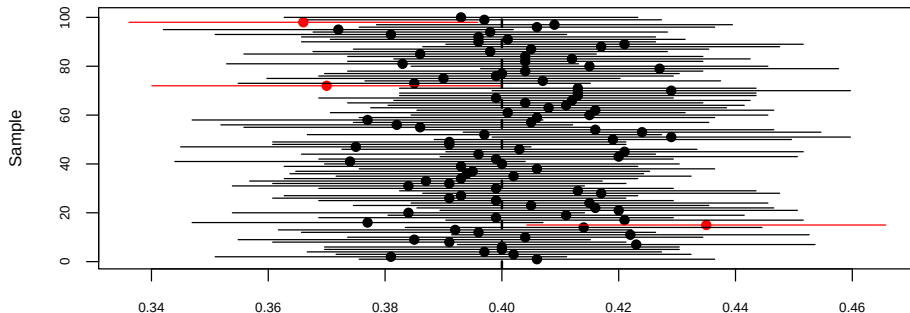
Do All Confidence Intervals Contain the Truth?

No. Some confidence intervals **miss the true population value**.

In the figure below:

- Black intervals **contain** the true value $p = 0.40$.
- Red intervals **do not contain** the true value.

95% Confidence Intervals from Repeated Samples (100 times)



The interpretation: if we repeated this *procedure* many times, 95% of the resulting intervals would contain the true parameter (the estimand) μ .

- The 95% is a property of the *procedure*, not of any one interval.
- This specific interval either contains μ or it doesn't—we just don't know which.

Today's Agenda

- Hypothesis Testing

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#) and from Brandon Stewart's [SOC 500 course materials at Princeton](#).

Hypothesis Testing

Today we flip the question

From “what values of the estimand θ are plausible?” to “is *this estimate* $\hat{\theta}$ plausible?”

Confidence interval: “Which values of the estimand θ are plausible given the data?”

Hypothesis test: “Is our estimate $\hat{\theta}$ surprising if the true parameter equals a particular value?”

Same machinery — CLT, standard errors, the sampling distribution.
Just packaged differently.

Example: Did Support Change After the Campaign?

Suppose we want to assess whether a campaign event increased voter support for a candidate in Pennsylvania.

We survey 2000 voters before and after the event. Code support as **1 = supports the candidate** and **0 = does not support the candidate**.

	Before the event	After the event
Sample mean	$\bar{Y}_{\text{before}} = 0.48$	$\bar{Y}_{\text{after}} = 0.52$

The observed difference is:

$$\hat{\theta} = \bar{Y}_{\text{after}} - \bar{Y}_{\text{before}} = 0.52 - 0.48 = 0.04$$

Support in our samples increased by **4 percentage points**.

Hypothesis Testing asks **If support in the population had not changed, how surprising would this sample difference be?**

Hypothesis testing follows the logic of a criminal trial

Trial	Hypothesis test
Defendant is innocent	H_0 : no effect
Prosecution argues guilty	H_1 : there is an effect
Evidence presented	Data collected
Beyond reasonable doubt?	Is $ t \geq c$? or Is $p \leq \alpha$?
Verdict: guilty / not guilty	Reject H_0 (Yes) / fail to reject H_0 (No)

- t : a measure of evidence against the null hypothesis.¹
- c : critical value
- p value: another measure of evidence against the null hypothesis.²
- α : significance level

¹the observed distance between the estimate and the null value, measured in standard-error units.

²the probability of observing a t statistics as extreme as the one observed, assuming the null hypothesis is true.

The key asymmetry

We assume innocence until proven guilty:

We never prove H_0 true — we only fail to reject it

- “**Not guilty**” $\neq$ “**innocent**” — insufficient evidence to convict
- “**Fail to reject H_0** ” $\neq$ “**no effect**” — insufficient evidence to reject
- “**No evidence of an effect**” $\neq$ “**evidence of no effect**”

This asymmetry is the most important idea in hypothesis testing

Reject H_0 or Fail to reject H_0

The null hypothesis is the claim we put on trial

Definitions

- **Null hypothesis (H_0):** the default claim (usually “no effect”)

Example: There **is not** a difference in change in voter support before and after our campaign.

- **Alternative hypothesis (H_1):** what we believe if we reject H_0

Example: There **is** a difference in change in voter support before and after our campaign.

Two-sided (standard):

$$H_0 : \theta = \theta_0 = 0 \text{ vs. } H_1 : \theta \neq \theta_0 \neq 0$$

Two-Sided and One-Sided Tests

A **two-sided test** considers departures in either direction:

$$H_0 : \theta = \theta_0 \quad H_1 : \theta \neq \theta_0.$$

A **one-sided test** considers a specified direction:

$$H_0 : \theta \leq \theta_0 \quad H_1 : \theta > \theta_0,$$

or

$$H_0 : \theta \geq \theta_0 \quad H_1 : \theta < \theta_0.$$

Choose the alternative based on the research question, before examining the results.

T Statistics and Null Distributions

A **t statistic** summarizes how far the sample estimate is from the value specified by H_0 , relative to its uncertainty:

$$T_n = \frac{\hat{\theta} - \theta_0}{SE(\hat{\theta})}.$$

- $\hat{\theta}$: the sample estimate.
- θ_0 : the hypothesized value under H_0 .
- $SE(\hat{\theta})$: the estimated standard error.

The test statistic varies across samples, so it is a **random variable**.

A **null distribution** is the sampling distribution of the test statistic **assuming that H_0 is true**.

T Statistics and Null Distribution

We know from the **Central Limit Theorem** that the standardized difference in means has a standard normal distribution asymptotically,

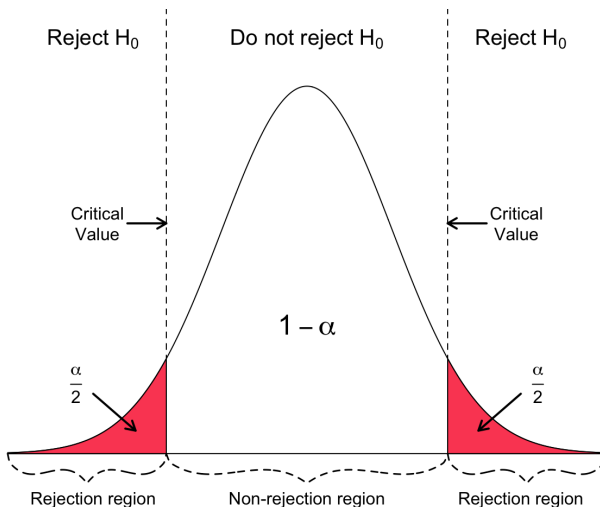
$$T_n = \frac{\hat{\theta} - (\mu_y - \mu_x)}{\widehat{\text{SE}}[\hat{\theta}]} \xrightarrow{d} \mathcal{N}(0, 1)$$

Under the null hypothesis of $\mu_y - \mu_x = 0$, we have

$$T_n = \frac{\hat{\theta}}{\widehat{\text{SE}}[\hat{\theta}]} \xrightarrow{d} \mathcal{N}(0, 1)$$

If T_n is very far from zero—in the sense that it has low probability under $\mathcal{N}(0, 1)$ —then we **reject** the null hypothesis as not plausible.

Two-Sided Rejection Region



Much like the 95% confidence interval, we pick $\alpha = .05$ by convention and have the critical value of 1.96.

t-statistics: Pennsylvania Voter Support

We test $H_0 : p_{\text{after}} - p_{\text{before}} = 0$ against $H_A : p_{\text{after}} - p_{\text{before}} \neq 0$.

The estimated difference and its standard error are:

$$\hat{\theta} = 0.52 - 0.48 = 0.04, \quad SE(\hat{\theta}) \approx \sqrt{\frac{0.52(0.48)}{2000} + \frac{0.48(0.52)}{2000}} \approx 0.0158.$$

The test statistic is:

$$t = \frac{\hat{\theta} - 0}{SE(\hat{\theta})} = \frac{0.04}{0.0158} \approx 2.53.$$

For a two-sided test at $\alpha = 0.05$, the large-sample critical values are approximately ± 1.96 .

Since $|t| = 2.53 > 1.96$, we **reject H_0** . There is evidence that population support changed.

From the t Statistic to the p-Value

So far, we have compared the **t statistic with a critical value**:

$$|t_{\text{obs}}| > \text{critical value} \Rightarrow \text{Reject } H_0.$$

This comparison uses **positions on the horizontal axis** of the null distribution.

We can also express the same evidence as an **area under the curve**: the probability of a result this extreme or more extreme, assuming H_0 is true.

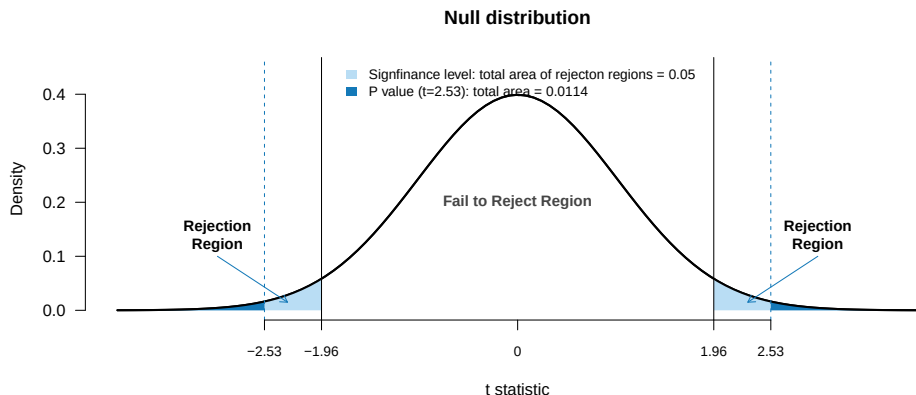
This area is the **p-value**, which we compare with the **significance level** α :

$$p \leq \alpha \Rightarrow \text{Reject } H_0.$$

The significance level (α) measures the size of the rejection region.

The two approaches give the same decision for the same test.

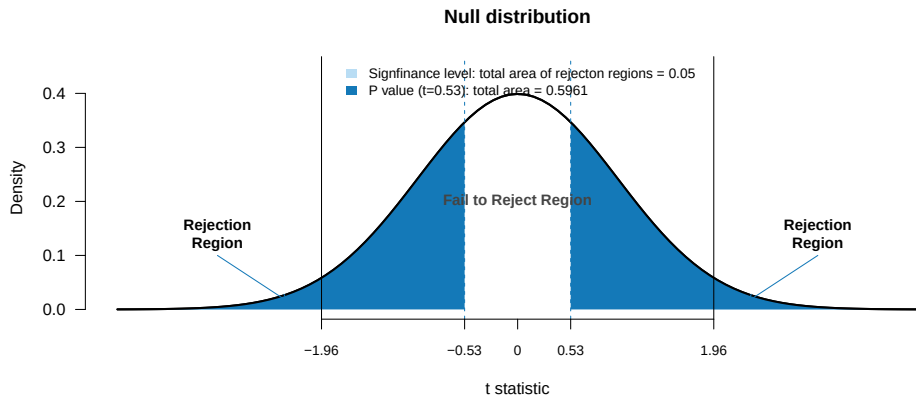
The p-Value and the Significance Level



- **P-value** = total area = $\mathbb{P}(|T| \geq |t| \mid H_0)$
- **Why $p/2$ on each side?** Two-sided test: a large t in *either* direction is evidence against H_0
- $N(0, 1)$ is symmetric $\Rightarrow$ each tail contributes exactly half: $p = p/2 + p/2$

Eg. when $t=2.53$, the p-value is smaller than α , so we reject H_0 .

The p-Value and the Significance Level



- Eg. when $t=0.53$, the p-value is larger than α , so we fail to reject H_0 .

p-value: Pennsylvania Voter Support

In our Pennsylvania example, $t_{\text{obs}} = 2.53$.

For a two-sided test:

$$p = P(|T| \geq 2.53 \mid H_0) \approx 0.011.$$

We can reach the same decision in two ways:

- **Critical-value approach:** $|t| = 2.53 > 1.96$.
- **p-value approach:** $p \approx 0.011 < 0.05$.

Both lead us to **reject H_0** .

p-value: In R

```
# Store the observed t statistic
t_obs <- 2.53

# Calculate the two-sided p-value:
# abs(t_obs)
# gives the distance from zero, regardless of sign.
# pnorm(-abs(t_obs))
# gives the area in the left tail below -2.53.
# Multiply by 2
# to include the equally large right tail above 2.53.
p_value <- 2 * pnorm(-abs(t_obs))

# Display the p-value (approximately 0.0114)
p_value

## [1] 0.01140625
```

When Do We Reject the Null Hypothesis?

For a **two-sided test**, the significance level α determines the critical value c .

Significance level α	Critical value c	Reject H_0 if
0.10 (10%)	1.645	$ t \geq 1.645$, or $p \leq 0.10$
0.05 (5%)	1.960	$ t \geq 1.960$, or $p \leq 0.05$
0.01 (1%)	2.576	$ t \geq 2.576$, or $p \leq 0.01$

For our result, $t = 2.53$ and $p \approx 0.0114$:

- At $\alpha = 0.10$: **reject** H_0 .
- At $\alpha = 0.05$: **reject** H_0 .
- At $\alpha = 0.01$: **fail to reject** H_0 .

A smaller α requires a larger $|t|$ and a smaller p to reject H_0 .

Three Things p-Values Do NOT Mean

WRONG: “ $p = 0.03$ means there is a 3% chance H_0 is true.”

Right: If H_0 were true, there would be a 3% chance of a result this extreme or more extreme.

WRONG: “ $p = 0.06$ means there is no effect.”

Right: The evidence is insufficient to reject H_0 at $\alpha = 0.05$. The effect might still exist.

WRONG: “ $p = 0.001$ means the effect is large.”

Right: A small p-value can result from a small effect with a large sample size.

The Full Hypothesis Testing Procedure

- 1 **State hypotheses:** H_0 and H_1 .
- 2 **Set the significance level:** $\alpha = 0.05$.
- 3 **Compute the test statistic:** $t = (\hat{\theta} - \theta_0)/SE(\hat{\theta})$.
- 4 **Find the p-value:** for a two-sided test, $p = \mathbb{P}(|T| \geq |t| \mid H_0)$.
- 5 **Make a decision:** reject H_0 if $p \leq \alpha$; otherwise, fail to reject H_0 .
- 6 **Interpret in context:** report the effect size and confidence interval.

Error Types

	H_0 False	H_0 True
Reject H_0	Correct	Type I error
Don't Reject H_0	Type II error	Correct

Suppose H_0 states that there is **no effect** ($\theta = 0$).

- H_0 is **false**: a real effect exists ($\theta \neq 0$).
- H_0 is **true**: no effect exists ($\theta = 0$).

We generally make the **normative** judgment that we prefer an **undetected finding** (Type II error) to a **false discovery** (Type I error).

Error Types: Does a Drug Work?

Suppose we test whether a new drug is more effective than a placebo.

- H_0 : the drug provides no additional benefit.
- H_A : the drug provides an additional benefit.

	Drug works (H_0 false)	Drug does not work (H_0 true)
Reject H_0	Correct	Type I error
Fail to reject H_0	Type II error	Correct

- **Type I error (false positive)**: conclude that the drug works when it does not.
- **Type II error (false negative)**: fail to detect a benefit when the drug actually works.

Hypothesis Testing in R

Let's move to

`R_Lab_W4_HypothesisTesting.R`

Wrap-Up (Week 3)

- Random Variable
- Probability distribution
- Expected value
- Variance and Standard deviation
- Covariance and Correlation
- Statistical Inference
- Estimand, Estimator, Estimate
- Random sampling
- Independent, Identically Distributed (IID)
- Sampling distribution

Wrap-Up (Week 4)

- Properties of Estimator
 - Unbiasedness
 - Efficiency
 - Variance of the Sample Mean
 - Standard Error
 - Mean Squared Error
 - Confidence Interval
 - Hypothesis Testing
 - Steps in Hypothesis Testing
 - t statistics ($|t|$ against critical value)
 - p value (against α)
 - State a conclusion (reject / fail to reject to H_0)
 - Error Types